

Analytics and Aesthetics in Sports

Scott Powers

September 16, 2026

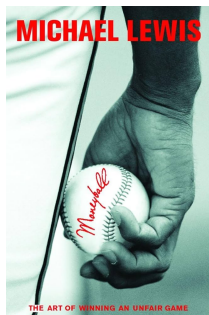


RICE UNIVERSITY
Sport Analytics

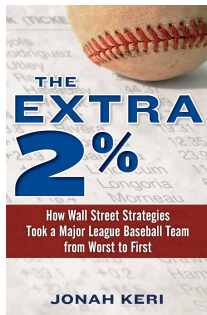
What is sport analytics?



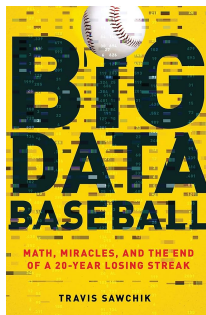
Books about baseball teams using analytics to win



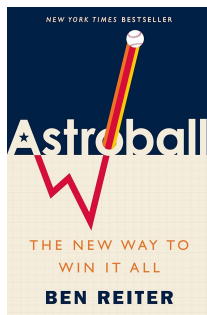
2002
Oakland
Athletics



2008
Tampa Bay
Rays



2013
Pittsburgh
Pirates



2017
Houston
Astros

MLB R&D staff sizes (according to Reddit)

Team	Analysts	Team	Analysts
American League		National League	
Yankees	20	Dodgers	20
Astros	15	Braves	15
Rays	15	Brewers	11
Angels	10	Reds	11
Tigers	9	Phillies	10
Rangers	8	Nationals	8
Mariners	7	Pirates	8
Royals	7	Padres	7
Twins	7	Cardinals	6
Blue Jays	6	Cubs	6
Red Sox	6	Giants	6
Indians	5	Marlins	6
Orioles	5	Dbacks	5
Athletics	3	Rockies	4
White Sox	2	Mets	3

2018

MLB Teams Ranked by Approximate R&D Size
Includes Data Scientists/Analysts, Data/Software Engineers, Biomechanists, etc.

TEAM	SIZE	TEAM	SIZE
	42		20
	41		20
	36		20
	35		20
	34		19
	34		19
	29		18
	28		17
	27		16
	26		16
	25		16
	24		15
	24		12
	23		12
	20		12
	20		10

Teams With No Clear Info: A's, Angels, Cubs, D-Backs

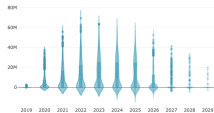
Amrit Vignesh | @avsportsanalyst

2025

What are they all doing?

Player Evaluation

Executives use modeling results to decide which players to target in trades and in free agency.



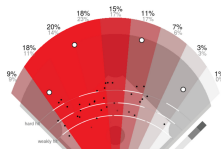
Player Development

Players use data-driven insights to tweak their game and achieve greater levels of performance.



In-Game Strategy

Coaches use analytics tools to position fielders and decide when to make substitutions.



The spectrum of sport analytics



Big moments in sport analytics

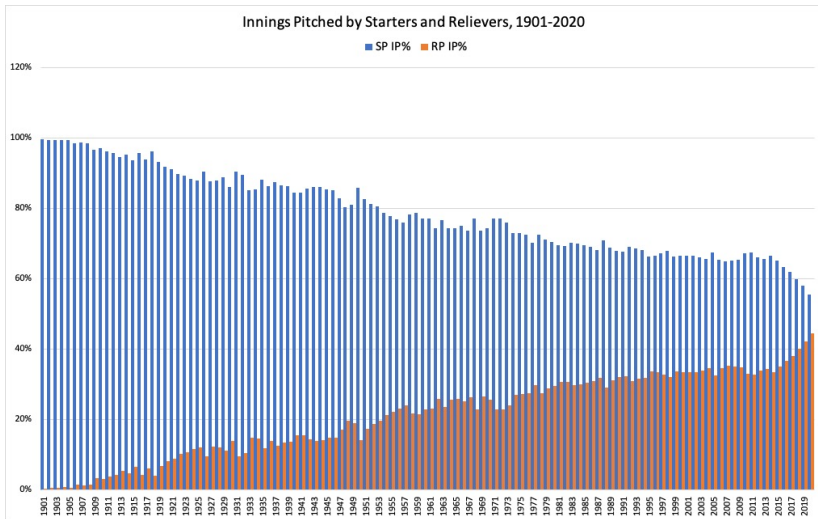
1. 2020 World Series Game 6
2. 2024 NBA Finals Game 1
3. Super Bowl LII

... and their implications for aesthetics

2020 World Series Game 6



A controversy 120 years in the making

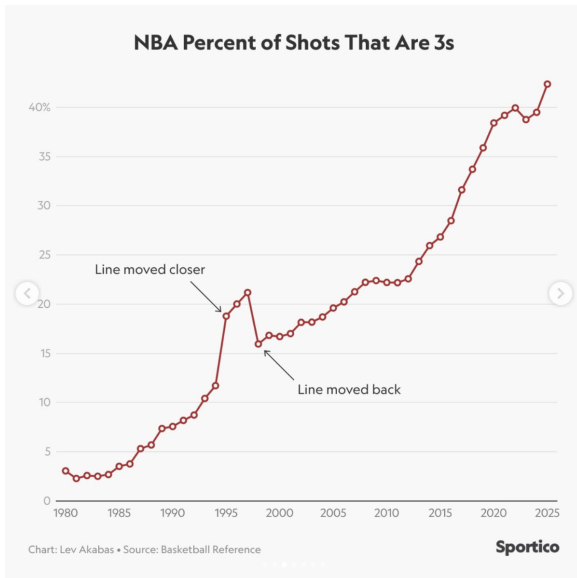


Bobby Mueller, *Off the Bench*

2024 NBA Finals Game 1



Three is greater than two

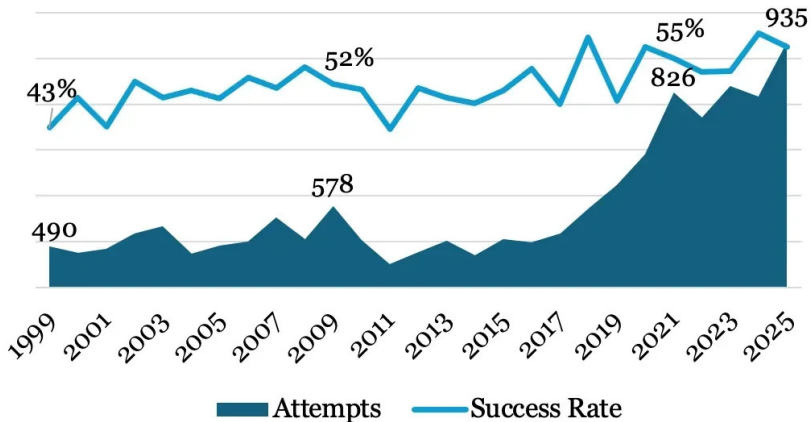


Super Bowl LII



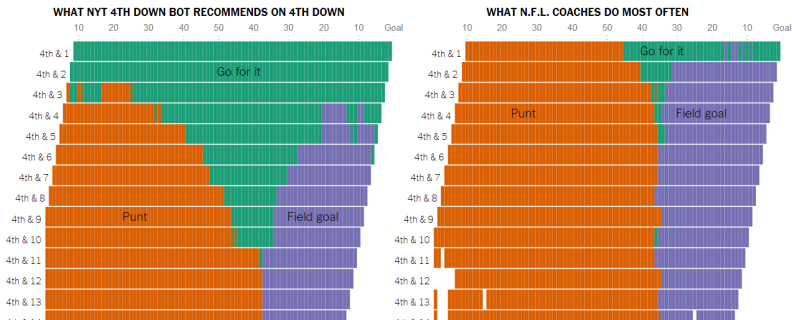
Go for it!

NFL Fourth Down Attempts And Success Rate, 1999-2025



https://www.reddit.com/r/nfl/comments/1raxi4n/oc_going_for_it_on_fourth_down_analytics_or/

Playing it by the book



New York Times 4th Down Bot

Summary

1. Teams use analytics to win more games through **player evaluation**, **player development**, and **in-game strategy**
2. Improved strategy is **sometimes less** aesthetically pleasing, **sometimes more** aesthetically pleasing
3. Some sports (e.g., baseball, volleyball) are more amenable to analysis than others (e.g., soccer)